PAPER_1408 — UQFF Derivation of Bertrand Probability Paradox (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Probability (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_paradox({'paradox':'bertrand_paradox'})` `→ 1/D_phys = 1/4 EXACT`

Observation

Three classical answers (1/2, 1/3, 1/4) for the probability that a random chord on a circle is longer than the inscribed equilateral triangle's side — paradox of arbitrary measure.

UQFF Closed Identity

```
P = 1 / D_phys = 1/4 EXACT (random-endpoint measure selected by F_U=1)
```

Physical Interpretation

Among the three measures, only the random-endpoint construction respects F_U=1 uniform normalization. The 1/D_phys = 1/4 answer is the structurally privileged one.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** {D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60} and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
